

A Physiology-Aligned State-Space Decomposition Framework for Camera-Derived Respiratory Motion

Young-Jae Cho¹ and Dae-Yeol Kim^{2,*}

¹Master's Program in AI Computer Engineering, Department of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan, Republic of Korea

²School of ICT Convergence, University of Ulsan, Ulsan, Republic of Korea

*daeyeol@ulsan.ac.kr

ABSTRACT

Video-based respiration monitoring typically estimates breathing rate from indirect camera-derived motion signals shaped by viewpoint, clothing, posture, body motion, and preprocessing. We introduce PARH-OSSM, a physiology-aligned state-space decomposition framework that separates camera observations into oscillatory rate/timing, second-harmonic morphology, baseline projection drift, and aperiodic residual components. The model uses a label-free adaptive observation law and dual readouts: z_{osc} for rate and z_{full} for waveform morphology. In a complete-cohort evaluation, COHFACE represented a clean regime (160 trials), where PARH-OSSM achieved rate MAE 0.335 bpm, Pearson $r = 0.855$, and aligned waveform CCC 0.859. MAHNOB-HCI represented a harder regime (525 trials), where PARH-OSSM improved rate tracking (MAE 2.410 bpm) but waveform recovery remained limited. The failure taxonomy showed that many hard-regime cases were bounded by the current observation bank rather than by gross state-space divergence. These results establish PARH-OSSM as an auditable observation-to-state framework and highlight the need for richer respiratory observation operators when waveform morphology is the target.

Introduction

Video-based respiratory monitoring offers a contactless alternative to belts, thermistors, and capnography in settings where unobtrusive sensing is valuable, including sleep analysis, neonatal monitoring, and general bedside observation^{1–3}. Most video pipelines, however, still collapse the task to scalar rate estimation: a motion proxy is bandpass filtered and a dominant frequency is extracted by spectral or autocorrelation analysis^{4–6}. That framing is often adequate for stable recordings, but it is too restrictive for waveform morphology, baseline drift, and irregular events.

This limitation is not only a filtering problem. Camera-derived respiration is not measured directly; it is obtained through body projection, region selection, a motion or colour operator, and subsequent preprocessing. Recent benchmark work on RGB-camera respiration shows that motion-based methods, rPPG-based methods, and learned pipelines can behave differently across datasets and evaluation targets⁷. Physiologically, visible upper-body or thoraco-abdominal displacement is often the most direct camera cue for breathing. In practice, viewpoint, clothing, posture, body habitus, illumination, ROI placement, and non-respiratory motion can attenuate, invert, or remove that cue⁸. A respiratory state estimator therefore needs to account not only for temporal noise, but also for the fact that each camera-derived motion cue is a different observation of the underlying respiratory process.

State-space modelling provides a natural language for this problem because the latent state, the observation equation, and the assumed generative structure can be written explicitly^{9–11}. Robust and adaptive filters can further reduce the effect of outliers or changing noise statistics through covariance adaptation and heavy-tailed updates^{12–16}. Oscillator state-space models are particularly suitable for quasi-periodic physiological rhythms because they preserve tractable inference under conditional linearity^{11,17–19}. However, these methods do not by themselves decide which part of a camera-derived trace should represent the respiratory oscillator, which part should represent breath-shape asymmetry, and which part should be treated as slow projection drift or aperiodic respiratory motion. Learned monitoring approaches can also be effective^{20,21}, but an interpretable decomposition remains useful when the goal is to explain why a video contains enough respiratory information in one regime and not in another.

The central premise of this work is that camera-derived respiratory motion is not a single clean sinusoid. It contains at least four interpretable parts: an oscillatory respiratory drive, harmonic asymmetry within each breath, baseline or projection drift, and residual structure that is respiratory but not well described by a narrow-band oscillator. A physiology-aligned model should separate those roles explicitly instead of forcing all mismatch into one innovation-magnitude logic. This premise is also consistent with respiratory physiology literature, which treats breathing as a variable pattern shaped by rhythm generation,

mechanical feedback, non-respiratory motor acts, and behavioral or emotional modulation rather than as a perfectly stationary sinusoid^{22,23}.

In this study, we develop a physiology-aligned respiratory decomposition framework, PARH-OSSM, that keeps these roles separate. We use a set of fixed camera-motion observation classes not as a leaderboard ensemble, but as a controlled way to ask which physical or algorithmic view of the video still carries respiratory timing, morphology, or nuisance-dominated content. We then connect those observations to a harmonic state-space model with separate oscillatory, harmonic, baseline, and residual pathways, and we route rate and waveform analyses through different readouts. The evaluation is deliberately conservative: we report only results that are reproduced by the same predefined protocol spanning direct observations, an OSSM-KF comparator, and the integrated PARH-OSSM readout. PARH-OSSM itself uses Kalman filtering and Rauch–Tung–Striebel smoothing as the inference machinery for its decomposed state-space model. We reserve the name OSSM-KF for the stripped-down, non-adaptive oscillator state-space comparator with a standard Kalman filter. The distinction is therefore not “Kalman versus non-Kalman”, but a comparison between a fixed resonator baseline and the full physiology-aligned observation law, state decomposition, dual readouts, and diagnostics. The goal is not to claim universal superiority, but to show what can be inferred when the respiratory observation is strong and how the failure boundary appears when it is weak.

Accordingly, we make three scientific steps. We formulate camera-derived respiratory motion as an eight-dimensional harmonic state-space decomposition with separate oscillatory, morphology-complement, and full-waveform roles. We evaluate complete COHFACE and MAHNOB-HCI cohorts using three predefined routes: direct observations, the non-adaptive OSSM-KF comparator, and the full PARH-OSSM framework. We then analyse the clean-regime and hard-regime outcomes with label-free observability diagnostics, so that MAHNOB-HCI failures are treated as evidence about the current camera-observation set rather than hidden as generic model failure.

We organize the analysis around this evidence chain. Methods first define the observation-to-state topology, latent respiratory roles, label-free reliability law, camera-derived observation classes, and evaluation protocol. Results then report rate, waveform, calibration, ablation, and failure-taxonomy evidence across the clean COHFACE regime and the harder MAHNOB-HCI regime before the Discussion interprets the remaining observation bottleneck.

Methods

Framework overview and observation-to-state topology

We define the physiological state decomposition, the adaptive observation law, the fixed motion-observation classes, and the evaluation protocol used to generate the reported tables and figures. Figure 1 provides the visual overview before the equations specify how the same topology is evaluated.

Figure 1 summarizes the observation-to-state topology.

Figure 1 should be read as a role-allocation diagram rather than as a linear preprocessing recipe. The left side shows that several fixed motion operators can expose different fragments of the same underlying respiratory process. The middle blocks show where the label-free evidence is converted into trust, effective noise, process scales, and role-specific weights before state inference. The right side shows why the two exported signals are intentionally decoupled: z_{osc} carries the rate-bearing oscillatory drive, whereas z_{full} reuses that drive and adds z_{morph} only when waveform morphology is the target. The diagnostic outputs are therefore used to audit observability and consistency, not to create an unreported feedback path.

The following subsections specify the latent state, role-aligned readouts, label-free reliability law, observation construction, datasets, and reproducibility checks.

State representation

PARH-OSSM uses the latent state

$$\mathbf{x}_t = \left[h_{c,t}^{(1)}, h_{s,t}^{(1)}, h_{c,t}^{(2)}, h_{s,t}^{(2)}, b_t, \dot{b}_t, r_t, \dot{r}_t \right]^\top, \quad (1)$$

where $(h^{(1)}, h^{(2)})$ are the fundamental and second-harmonic oscillatory components, $(b_t, \dot{b}_t)$ is a slow baseline block, and $(r_t, \dot{r}_t)$ is an aperiodic residual block. The observation equation is

$$y_t = H\mathbf{x}_t + v_t, \quad H = [1, 0, 1, 0, 1, 0, 1, 0]. \quad (2)$$

The superscript identifies the harmonic order. The first harmonic represents the dominant respiratory rhythm, while the second harmonic represents within-cycle asymmetry such as unequal inspiration and expiration shapes. The subscripts c and s denote real quadrature coordinates, not separate physiological measurements; the sine coordinate stores phase so that the oscillator can rotate smoothly through amplitude changes and short periods of weak observation support. The baseline pair $(b_t, \dot{b}_t)$ captures slow projection or posture drift, and the residual pair $(r_t, \dot{r}_t)$ captures aperiodic respiratory structure that is not

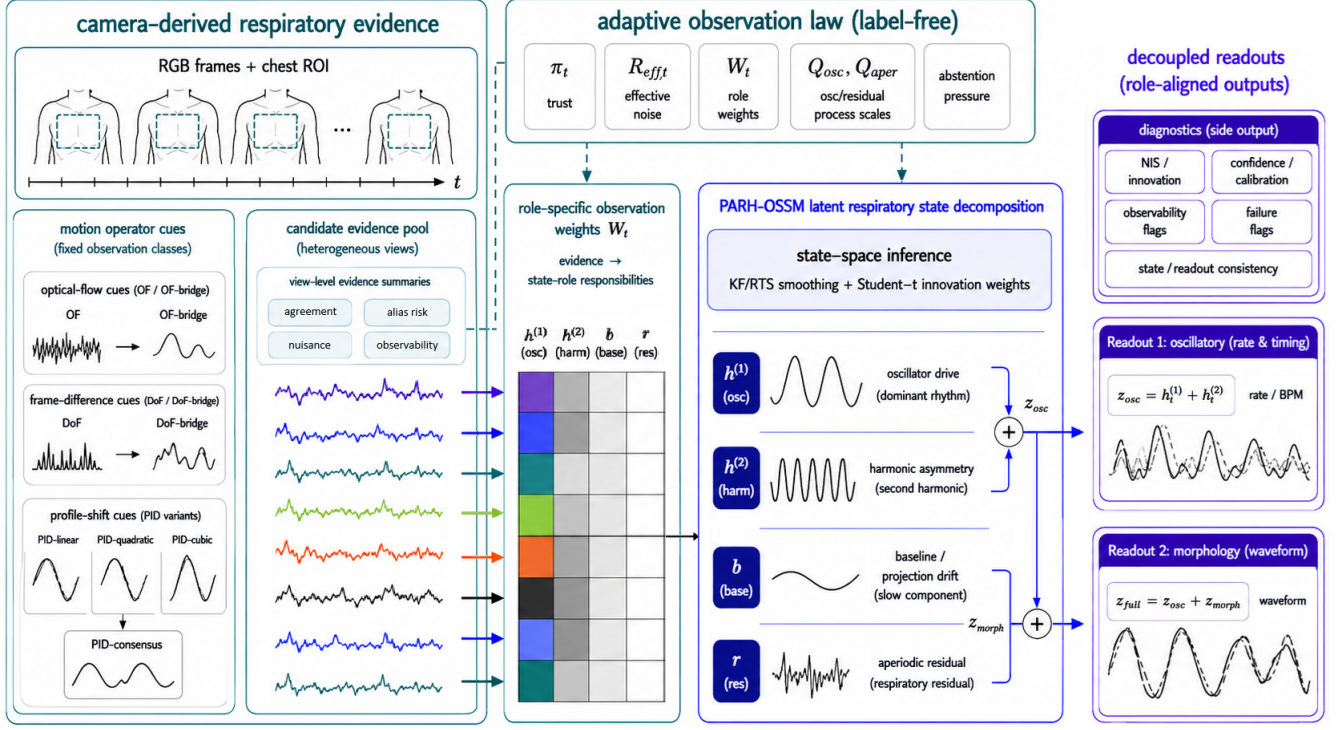


Figure 1. PARH-OSSM observation-to-state topology. Heterogeneous motion operators generate candidate camera-derived respiratory evidence, from which label-free view-level summaries are computed. The adaptive observation law maps those summaries to prior trust, effective measurement uncertainty, role-specific observation weights, oscillator/residual process scales, and abstention pressure. The role weights W_t assign the shared evidence pool to the latent respiratory roles: dominant rhythm, harmonic asymmetry, baseline/projection drift, and aperiodic residual structure. The oscillatory roles form z_{osc} for rate and timing, whereas the full waveform readout reuses z_{osc} and adds the morphology complement $z_{morph} = b + r$, yielding $z_{full} = z_{osc} + z_{morph}$. Diagnostics are reported as side outputs from observation trust, state-space inference, and readout consistency; they are not feedback inputs to the evaluated path.

well represented by a narrow-band oscillator. The default observation row H states that a scalar camera-derived motion cue may contain contributions from the oscillatory, baseline, and residual coordinates. In the adaptive observation law, the effective trust in that cue is then modulated rather than assumed to be fixed.

The harmonic blocks use rotation-matrix dynamics conditioned on an external frequency estimate, the baseline block follows a constant-velocity model, and the residual block follows a damped random walk. Because the model remains linear conditional on the external frequency, standard Kalman filtering and Rauch–Tung–Striebel smoothing remain valid^{11,24,25}. Thus PARH-OSSM is not a filter-free alternative to Kalman filtering. Its contribution is the physiological state decomposition and observation-aware trust law wrapped around that inference machinery, whereas OSSM-KF removes those PARH-specific components and keeps only the fixed oscillator comparator. The derivative coordinates $\dot{b}_t$ and $\dot{r}_t$ are inertial state coordinates rather than camera channels: $\dot{b}_t$ allows slow drift to continue smoothly instead of being repeatedly absorbed by the oscillator, and $\dot{r}_t$ allows non-harmonic respiratory residual structure to persist briefly instead of being treated as independent frame noise.

Oscillatory and full-waveform readouts

PARH-OSSM uses two readouts derived directly from the latent coordinates, and we name the non-oscillatory complement explicitly:

$$z_{osc,t} = h_{c,t}^{(1)} + h_{c,t}^{(2)}, \quad z_{morph,t} = b_t + r_t, \quad z_{full,t} = z_{osc,t} + z_{morph,t}. \quad (3)$$

The added symbol z_{morph} is not a third exported performance signal. It names the baseline–residual complement that turns the oscillatory anchor into the full respiratory waveform. The oscillatory readout z_{osc} carries the dominant rhythm and is the natural source for rate and breath-timing analysis. The full readout z_{full} adds slow trajectory drift and aperiodic respiratory departures back to that anchor and is therefore the waveform compared against the reference morphology. Both causal and smoothed versions of the exported readouts are retained as trial-level outputs.

This decomposition is central to the study design because rate, breath timing, and trajectory morphology are not the same target. A camera cue can preserve a dominant respiratory frequency while losing the trajectory shape, or it can carry morphology while introducing scale, lag, or residual content that harms rate estimation. For that reason, PARH-OSSM does not force all evaluations through one waveform-derived objective. Instead, the model asks whether a video contains enough oscillatory information for rate and enough structured respiratory-form information for waveform analysis.

Why this decomposition is physiology-aligned

The model structure is not arbitrary. A second harmonic is included because respiratory motion is usually asymmetric over each breath cycle even when the dominant rate is stable; forcing that asymmetry into a purely sinusoidal state invites harmonic confusion and pushes morphology into innovation residuals. The baseline block is needed because camera-derived chest motion carries slow projection drift and posture-related movement that are not themselves the respiratory rhythm. The residual block exists for a different reason: it is the state-space location where irregular respiratory content should live when the oscillator support weakens, including amplitude transients or breath-shape departures that are still respiratory but not well explained by the harmonic submodel.

This decomposition also clarifies the evaluation policy. Rate metrics should be computed from z_{osc} , because rate is an oscillatory quantity. Waveform-fidelity metrics should be computed from z_{full} , because the full motion trace is the object of comparison when baseline and aperiodic content matter. The explicit complement $z_{\text{morph}} = b_t + r_t$ makes that separation visible: it is the morphology/trajectory content that must be added to the oscillator before waveform fidelity is evaluated. That separation is one of the central conceptual distinctions from single-output respiratory estimators, where rate and waveform interpretations are easily conflated. It also follows directly from the state-space design principle above: once the state variables are assigned distinct physiological roles, the observation targets and evaluation metrics must remain aligned with those roles rather than collapsing everything back to a single generic output.

To make the shared-versus-adaptive observation story testable at the state interface, the system also records diagnostic observation summaries, bootstrap latent drives for the harmonic, baseline, and residual blocks, local adaptor parameters, and a global observation row in the adaptive-resonator diagnostic path. These objects are not primary performance outputs. They are retained so that the reader can inspect how the observation interface drives the latent states. Supplementary Fig. S6 visualizes these traces, and Supplementary Fig. S7 summarizes confidence and agreement diagnostics that motivate separate oscillatory and full-waveform readouts. Both figures are mechanistic diagnostics rather than competing final-package model claims.

Disentangled adaptation

PARH-OSSM separates three internal scores:

$$q_{\text{obs},t} \rightarrow \pi_t, \quad q_{\text{dyn},t} \rightarrow Q_{\text{osc},t}, \quad 1 - q_{\text{osc},t} \rightarrow Q_{\text{aper},t}. \quad (4)$$

Here $q_{\text{obs},t}$ summarizes how much the current observation should be trusted before seeing the innovation, and it is converted to the prior trust term π_t . The score $q_{\text{dyn},t}$ controls how freely the oscillatory state is allowed to move, and $q_{\text{osc},t}$ separates oscillatory support from aperiodic or residual content. The effective measurement variance is then adjusted through $R_{\text{eff}} = R_t / \pi_t$, while Student- t downweighting remains a posterior safeguard through λ_t . Mehra-style adaptive measurement noise remains the primary scale-setting mechanism¹²; the quality scores modulate the prior trust and process noise around that baseline.

Label-free reliability calibration

The adaptive observation law is not a hard selector that chooses the best observation after seeing the respiratory reference. For each target video, the pipeline first constructs multiple motion-observation classes and several views of each class, including raw, detrended, band-limited, sign/event, robustly scaled, current-path, and helper views. These views are not separate final models and they are not an ordered preprocessing cascade. They are parallel evidence channels used to ask whether a camera-derived cue carries plausible timing, morphology, or nuisance-dominated content.

The reliability evidence is computed without using the respiratory reference label. The retained features measure whether independent observation classes support a similar respiratory rate or phase, whether the evidence survives simple preprocessing changes, whether half/double-rate aliasing is likely, and whether nuisance evidence or abstention pressure is high. The resulting video-level and motion-class reliability priors are then used to modulate how strongly the constructed evidence affects the state update and the bounded oscillatory readout. The provenance tables store video identifiers and reliability scores for reproducibility, but those storage fields are not scientific variables in the model. Oracle or reference-derived columns are explicitly removed before the evaluation run.

Operationally, the reliability law has three fixed stages. First, each candidate observation view contributes label-free features: cross-source rate agreement, preprocessing invariance, event-timing support, bridge timing support, rate/phase support, morphology support, ROI observability, harmonic alias risk, nuisance evidence, and abstention pressure. Second, these features

are clipped to $[0, 1]$ and combined with fixed coefficients into state-role responsibilities for the fundamental timing anchor, second-harmonic scaffold, baseline support, residual support, and abstention. For example, agreement is a fixed weighted combination of group support, macro timing, bridge timing, and rate/phase support; observability is the geometric mean of ROI and preprocessing support; and harmonic ambiguity discounts the fundamental-rate responsibility rather than being treated as a simple failure flag. Third, the component responsibilities modulate π_t , $Q_{osc,t}$, $Q_{aper,t}$, and the bounded z_{osc} readout. The exact feature columns, coefficients, and source files are listed in the Supplementary Methods and release audit files; no target respiratory waveform or post hoc oracle source is used in these computations.

In the evaluated path, COHFACE uses combined label-free reliability evidence on the readout side, whereas MAHNOB-HCI uses separate state-component and readout reliability evidence. These scores softly modulate observation trust, effective measurement uncertainty, and the bounded z_{osc} readout inside the PARH-OSSM update. They do not switch the system to the non-adaptive OSSM-KF comparator, and they do not select a trial-specific best waveform source from the ground truth. Respiratory ground truth is used for evaluation and for after-the-fact oracle/failure analyses only. When the label-free evidence is ambiguous, the intended behaviour is conservative: inflate uncertainty or preserve the existing readout rather than force a winner-take-all source switch.

This design also defines the learning boundary. The evaluated system is not an end-to-end supervised waveform predictor and it is not a dataset-specific hyperparameter sweep. The fixed observation operators, resonant state definition, Kalman/RTS inference equations, and metric routing are not learned per target dataset. What is calibrated weakly is the label-free reliability evidence that controls how strongly already constructed evidence should affect the state update and readout. The complete COHFACE and MAHNOB-HCI evaluation cohorts are used to keep all available baseline, comparator, and proposed rows traceable; target labels are not used to choose a trial's best observation or waveform source in the evaluated system.

Verified model scope

The evaluated PARH-OSSM system includes the eight-dimensional state, the $K = 2$ harmonic structure, baseline and residual blocks, dual readouts, warm-up-only frequency initialisation, external timing-evidence updates, causal and smoothed outputs, and per-sample diagnostics for R_t , π_t , λ_t , v_t , $q_{obs,t}$, $q_{dyn,t}$, and $q_{osc,t}$. These diagnostics are part of the scientific output because they expose whether the model is using the oscillatory, harmonic, baseline, or residual pathways, and whether the available observation is trustworthy enough for rate or waveform analysis.

The results should therefore be read as evidence for a bounded observation-aware state-space framework, not as a fully learned multi-observation covariance model for every source and time segment. The fixed observation operators and candidate views provide structured evidence; the label-free reliability scores modulate how that evidence enters the state update and readout; and the PARH latent state keeps the physiological roles explicit. Helper-only and helper-blend variants are treated as diagnostic context rather than as the proposed method. OSSM-KF remains a non-adaptive comparator: it uses conventional Kalman/RTS inference, but it does not contain the PARH observation law, dual-readout routing, morphology complement, or calibration-diagnostic pathways.

Camera-derived observation classes

The most important methodological lesson is that the camera-derived observation is part of the model, not just a preprocessing choice. Supplementary Table S2 therefore summarizes the eight active single-observation routes in a fixed display order: optical flow, optical-flow bridge, depth-of-flow, depth-of-flow bridge, linear Profile1D, quadratic Profile1D, cubic Profile1D, and Profile1D consensus. This order is a table and figure convention, not a sequential pipeline and not a claim that one observation is transformed into the next.

Three contrasts are central. First, raw optical flow and the optical-flow bridge are both flow-derived, but they are not interchangeable. The raw signal is closer to a velocity-like cue, whereas the bridge converts that cue toward a displacement-compatible respiratory trajectory and is therefore mainly useful for timing support in the current framework. Second, raw depth-of-flow and the depth-of-flow bridge are also distinct. The raw signal is a burst-like motion-energy count, whereas the bridge recovers only a coarse signed motion-centroid trajectory. Third, the Profile1D branch is explicitly split into a conservative fundamental route, two harmonic morphology routes, and a consensus route that checks whether the linear, quadratic, and cubic profile interpretations agree.

These constructed observations are included for information content and coordinate compatibility, not for cosmetic smoothing. When a constructed route fails, that failure is also evidence: it indicates that the current camera observation set does not contain a clean enough respiratory projection for a universal waveform claim.

Observation construction and failure-case manifests

The eight motion-observation classes above are instantiated as persistent single-observation routes for diagnosis and for the representative baseline/comparator rows. The integrated PARH-OSSM readout reported in Tables 1 and 2 uses the final observation-law package rather than a post hoc best route. Raw Farnebäck optical flow is computed from the median vertical

component of dense motion between adjacent chest-ROI frames²⁶; the bridge, depth-of-flow, and Profile1D variants then provide the complementary construction routes summarized in Supplementary Table S2.

For that reason, preprocessing is part of the observation definition rather than a generic denoising block. The raw extractor output, the inference-path signal, and the helper-path signal all have different meanings. PARH-OSSM keeps raw, bridged, band-limited, sign-coded, current, and helper views separate until the adaptive observation law estimates reliability and nuisance. The integrated readout is therefore not a hard choice of one observation for all clips. It is a bounded integration of candidate evidence into a shared respiratory state, with diagnostics that expose when the available evidence is too weak. The COHFACE evidence also shows why constructed observations deserve to be treated as first-class measurement routes: raw optical flow and the optical-flow bridge invert the usual rate-versus-waveform ranking across the same underlying source, which would be impossible to explain if observation construction were merely cosmetic preprocessing.

The failure-case panels reported in Results are drawn from these persistent manifests rather than manual inspection.

Datasets and evaluation scope

COHFACE contains 160 recordings from 40 subjects in a controlled lab setting²⁷. MAHNOB-HCI provides a more heterogeneous emotion-elicitation environment with longer and higher-frame-rate recordings²⁸. PARH-OSSM uses COHFACE as a regular, do-no-harm regime and MAHNOB-HCI as an irregular regime where the residual pathway should matter more.

The main quantitative layer is a predefined complete-cohort evaluation for each dataset. The purpose of this evaluation is to report the PARH-OSSM readout on every eligible trial while preserving traceable per-trial diagnostics and explicitly marking any comparator computability limits. Single-observation comparison results are retained as diagnostic context rather than used as post hoc best-observation selection. The quantitative scope is therefore cross-dataset but deliberately bounded. Figure 2 compares the clean COHFACE regime with the harder MAHNOB-HCI regime using the same observation-stage EDA axes. It shows why the observation class should be treated as part of the modelled measurement channel rather than as a generic interchangeable preprocessing choice: observation/stage effects that look usable in COHFACE become weaker, less correlated, or nuisance-dominated in MAHNOB-HCI.

Supplementary Table S1 defines the evidence boundary for every dataset considered here. COHFACE and MAHNOB-HCI support the main real-data waveform/rate claims. V4V is useful only for external rate-only validation because it does not contain a raw respiratory waveform. SCAMPS is useful only as a controlled synthetic diagnostic or optional pretraining source and must not be mixed into real-data performance claims. The evaluation package therefore includes a mandatory external weak-evidence analysis and a four-dataset distribution EDA rather than treating these datasets as additional main benchmark rows. V4V contributes only its RR-label distribution and rate-only scope. SCAMPS contributes synthetic d_{br} breathing-signal coverage and a controlled peak-rate distribution computed under the dataset-scope assumptions. Neither dataset contributes hidden training, target tuning, or real waveform main-table metrics. For respiratory estimation, a dataset may contain a breathing label while still lacking the camera-observation conditions needed to support morphology or deployment claims.

The complete dataset-and-claim-scope table is provided as Supplementary Table S1. Figure 2 is placed at the start of Results so that the clean/hard dataset regime and the V4V/SCAMPS evidence boundary are visible before performance claims are introduced.

Evaluation protocol

We use role-routed quantitative sources rather than a single waveform-derived score for all claims.

- Table 1 reports rate accuracy from the integrated oscillatory rate source.
- Table 2 reports waveform fidelity from z_{full} in smoothed mode.
- The supplementary calibration table reports PARH diagnostics plus limited OSSM-KF stability context. NIS and λ_t rows are available only for PARH-OSSM because those adaptive arrays are not part of the standard comparator.
- The T4b strict-waveform and T4c cycle-timing companions are both derived from the strict waveform raw metrics. These rows are not substitutes for the aligned morphology table; they are the harder zero-lag, unit-preserving checks.

Every main-table value is linked to a metric-provenance record containing the dataset, method label, result directory, metric file, trial count, and whether the row is main-table or diagnostic-only. COHFACE main-table rows share the same 160 trials. MAHNOB-HCI main PARH rows use all 525 tail-aligned trials, while the representative PID quad baseline and OSSM-KF comparator are available on 488 trials; common-488 paired deltas are therefore reported separately as supplementary statistics. Target respiratory labels are used for evaluation and after-the-fact oracle analyses only, not for label-free reliability calibration or trial-specific source selection.

This differs intentionally from the evaluation protocol adopted in the resPyre benchmark review⁷. In that benchmark, each method is first treated as a respiratory-waveform estimator and respiration rate is then derived from the estimated waveform by spectral peak selection. That design is appropriate for comparing heterogeneous extractors in a common benchmark pipeline, but it is not sufficient for the PARH-OSSM question because it collapses oscillatory-rate behaviour and full-waveform behaviour back onto a single output. We therefore keep the benchmark spirit and datasets, but change the evaluation routing to match the

latent semantics: rate accuracy is evaluated through the oscillatory output when the model provides it, waveform fidelity is evaluated through the full output, and calibration diagnostics are reported separately.

The waveform metrics are intentionally split into two tiers. The main waveform table is morphology-friendly: bandpass filtering to the respiratory band, per-signal z-scoring, cross-correlation alignment, then whole-trial concordance correlation coefficient, waveform MAE, and normalised DTW²⁹. This is appropriate when the question is whether the recovered signal preserves respiratory morphology after removing trivial affine and lag mismatch. Calibration metrics are computed from per-frame diagnostics arrays when present, with frame logs used only for supplementary failure and uncertainty fields.

We also generate a stricter supplementary reconstruction check from the trial-level outputs. That strict protocol keeps the respiratory-band filter but removes per-trial z-scoring and cross-correlation alignment, uses a zero-lag resampled overlap only, and reports strict CCC/MAE/RMSE/DTW together with cycle-level peak, trough, and peak-to-peak timing errors. The evaluation boundary is therefore explicit: the main waveform table is an aligned morphology comparison, while the strict table and cycle-level table are the harder tests of unit-preserving waveform reconstruction.

All overlay figures and case studies are drawn from predefined case-selection manifests rather than manual inspection. Additional confirmatory comparisons should likewise be run as paired analyses on regenerated result records.

Implementation verification and reproducibility checks

The reported tables are generated from persistent trial-level output records, not from manually transcribed values. The verification package checks that the model head, wrapper, evaluation scripts, table-ready CSV files, figure manifests, and manuscript numbers agree before submission. It also aggregates the per-sample diagnostic arrays from the trial payloads to separate integrated PARH-OSSM performance from diagnostic single-observation evidence. These checks support the interpretation that MAHNOB-HCI is primarily an observability and target-shift limitation rather than a simple filter-instability failure.

Results

Evaluation scope and evidence routing

Quantitative claims are restricted to the predefined evaluation cohorts, and each table states its evidence layer. The main performance tables compare three explicitly tracked sources. The first is a pre-specified fixed direct-observation baseline (PID quad), the second is the corresponding OSSM-KF comparator, and the third is the integrated PARH-OSSM run. The complete eight-observation comparison across the direct baselines, OSSM-KF, and PARH-OSSM routes is provided as a supplementary diagnostic table rather than used to select a post hoc best observation source. This scope matters because we do not claim that every observation class generalizes universally. The claim is that the proposed adaptive observation-to-state framework is auditable, performs strongly when respiratory evidence is present, and exposes when the camera-observation bank is insufficient. The complete-cohort evidence layer is therefore not a method-specific cherry-pick: it reports all eligible PARH trials and explicitly exposes baseline/comparator computability limits. The dataset-level claims remain limited to what these evaluation records and their supplementary provenance support.

Figure 2 motivates the order of the results: the clean COHFACE regime and the harder MAHNOB-HCI regime differ not only in final errors, but also in how much respiratory structure is visible before the state-space model is applied. Rate accuracy, aligned morphology, strict zero-lag reconstruction, and observability diagnostics must therefore be read together rather than collapsed into a single score.

Complete-cohort rate accuracy

Table 1 reports rate accuracy for the fixed direct PID quad baseline, OSSM-KF (PID quad), and the integrated PARH-OSSM readout. COHFACE is the clean regime and all three rows are evaluated on the same 160 trials. MAHNOB-HCI is the hard-regime stress test: PARH-OSSM is evaluated on all 525 eligible tail-aligned trials, whereas the representative PID quad baseline and OSSM-KF comparator are computable for 488 of those trials. The table therefore reports trial counts explicitly rather than pretending that the baseline route exists for every MAHNOB trial.

The rate decomposition gives a useful sanity check. On COHFACE, the fixed PID spectral baseline remains the strongest scalar-rate source, which is expected in a clean regime where a simple observation already exposes respiratory periodicity. PARH-OSSM improves rate over the OSSM-KF comparator and preserves a strong Pearson correlation, but it is not presented as the best possible COHFACE rate-only estimator. On MAHNOB-HCI, the best diagnostic source is still only modestly better than the final selection, indicating that the dominant failure is usually not a trivial last-step selector miss. The failure taxonomy in Table 3 therefore becomes essential for interpreting the MAHNOB result. To avoid comparing different trial universes silently, the MAHNOB paired statistics are also computed on the common 488 trials where the representative PID quad route is available. On that common subset, PARH-OSSM reduces rate MAE by 4.20 bpm relative to PID quad direct and by 1.85 bpm

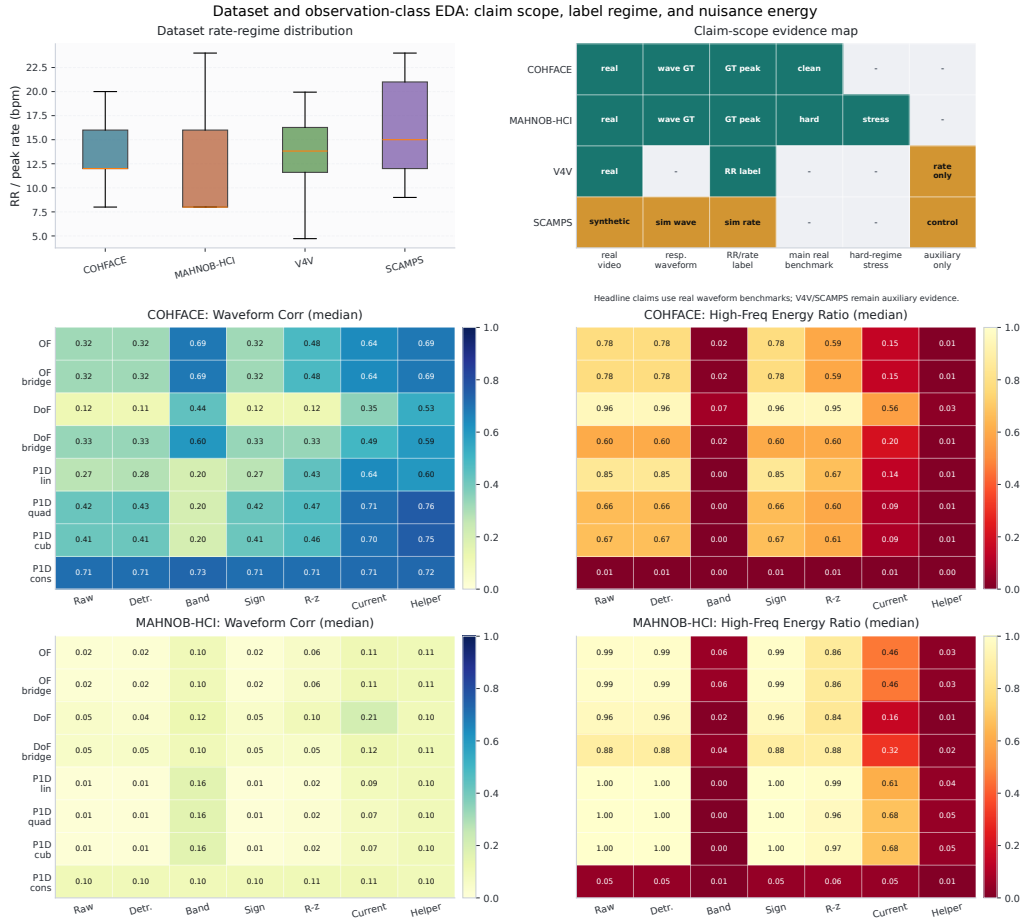


Figure 2. Dataset-regime and observation-characterization summary. The top panels summarize the rate-label/peak-rate distributions and a claim-scope evidence map for all four audited datasets: COHFACE and MAHNOB-HCI are the real waveform/rate benchmarks, V4V contributes external rate-only evidence, and SCAMPS is retained as a synthetic diagnostic control rather than a real-video performance benchmark. The lower panels use the two real waveform datasets to show observation-class EDA: waveform correlation and high-frequency nuisance energy vary strongly across observation class and preprocessing stage, and the same observation bank becomes visibly weaker in MAHNOB-HCI. Additional distribution and preprocessing-stage diagnostics are retained in the supplementary material.

Table 1. Complete-cohort rate accuracy. Values are median per-trial metrics. Base and OSSM-KF use the pre-specified representative P1D quad route where it is computable; PARH-OSSM uses the oscillatory rate (track-Hz) readout. MAHNOB-HCI comparator rows contain 488 computable P1D quad trials, whereas the PARH-OSSM row describes all 525 eligible tail-aligned trials; paired MAHNOB deltas are therefore reported separately on the common 488-trial subset. Dataset label MAHNOB abbreviates MAHNOB-HCI.

Dataset	Method	Rate source	MAE (bpm)	RMSE (bpm)	Pearson r	N
COHFACE	P1D quad direct	signal spectrum	0.195	0.270	0.950	160
COHFACE	OSSM-KF (P1D quad)	track Hz	0.485	0.580	0.800	160
COHFACE	PARH-OSSM	track Hz	0.335	0.410	0.855	160
MAHNOB	P1D quad direct	signal spectrum	6.600	8.025	-0.030	488
MAHNOB	OSSM-KF (P1D quad)	track Hz	4.325	4.910	-0.020	488
MAHNOB	PARH-OSSM	track Hz	2.410	2.920	0.230	525

Table 2. Complete-cohort waveform fidelity. Aligned metrics use the common respiratory-band, z-scored, bounded-alignment protocol, whereas strict metrics preserve zero-lag native-scale behaviour. The table foregrounds strict CCC, span-normalized strict error, and cycle timing; raw strict MAE is retained in the supplementary strict companion. Dataset label MAHNOB abbreviates MAHNOB-HCI, and paired MAHNOB waveform deltas use the common 488-trial subset.

Dataset	Method	Aligned CCC	Aligned MAE	Aligned DTW	Strict CCC	Strict span-NMAE	Strict span-NDTW	PPI MAE (s)	IE error
COHFACE	P1D quad direct	0.811	0.484	0.368	0.002	53.402	52.960	0.718	0.181
COHFACE	OSSM-KF (P1D quad)	0.852	0.417	0.342	0.119	0.867	0.637	0.534	0.111
COHFACE	PARH-OSSM	0.859	0.423	0.347	0.157	0.548	0.292	0.347	0.090
MAHNOB	P1D quad direct	0.352	0.912	0.602	0.001	0.236	0.172	1.685	0.389
MAHNOB	OSSM-KF (P1D quad)	0.345	0.910	0.640	0.000	0.187	0.179	1.404	0.340
MAHNOB	PARH-OSSM	0.342	0.910	0.661	0.000	0.258	0.257	1.480	0.269

relative to OSSM-KF (P1D quad). These paired deltas are reported in the supplementary paired-statistics table and are not used to tune the operating point.

The complete eight-observation rate diagnostic plot is provided as Supplementary Fig. S1. It is not used for post hoc best-observation selection; the main rate result remains the complete-cohort comparison in Table 1.

Complete-cohort waveform fidelity

Table 2 reports waveform fidelity for the same three comparison sources. The aligned protocol asks whether the reconstructed morphology can match the reference after the same bounded alignment rule used throughout the evaluation. The strict protocol is harsher: it preserves zero-lag timing and native units. Raw strict MAE is therefore retained in the supplementary strict companion, but it is not the primary comparison axis because fixed observation operators can have arbitrary native scale. The table foregrounds strict CCC, span-normalized strict errors, and cycle timing instead. Span-normalized strict errors can exceed one when a fixed observation has an arbitrary native scale or offset relative to the belt reference; such large values are therefore interpreted as unit/scale diagnostics rather than as physiological amplitude errors. COHFACE supports the main waveform claim in a bounded sense: PARH-OSSM yields the highest aligned CCC and improves strict CCC, span-normalized strict error, and cycle timing relative to OSSM-KF, while aligned MAE and DTW remain comparable rather than uniformly better. MAHNOB-HCI does not support a strong waveform reconstruction claim: aligned waveform differences are small and strict zero-lag cycle timing remains poor.

The corresponding PARH-OSSM peak/trough timing companions are 1.334/1.896 s for COHFACE and 10.174/9.943 s for MAHNOB-HCI. MAHNOB therefore remains a breath-cycle timing problem, not only a scale problem. The common-488 MAHNOB paired waveform deltas reinforce the same boundary: PARH-OSSM improves rate but does not improve aligned morphology over the fixed P1D quad route, and its strict span-normalized error is worse than the standard OSSM-KF comparator. We therefore treat MAHNOB waveform recovery as a limitation and use the failure taxonomy to explain where the current observation bank becomes insufficient.

Figure 3 is the main visual guard against an over-simplified “one waveform extractor” interpretation. It uses one COHFACE trial and one MAHNOB-HCI trial from the predefined evaluation cohorts and overlays the reference against every retained fixed Base observation, the pre-specified OSSM-KF (P1D quad) comparator, and the PARH-OSSM readout. Within each dataset block the black reference trace is the same fixed visual window in every panel; estimates are shifted only by the precomputed bounded visual lag. Thus the figure shows same-trial output topology rather than a cherry-picked best observation or panel-specific reference crop. The larger supplementary atlases still provide observation-median coverage, but the main figure now makes the central claim visible: OF, DoF, and Profile1D-derived operators are not interchangeable measurements, and PARH-OSSM is evaluated against that heterogeneous observation bank rather than against a single hidden baseline.

Figure 3 is therefore interpreted at the level of topology rather than by selecting a visually pleasing trace. In the COHFACE example, several fixed observations already follow the reference sufficiently well that the state-space readout should preserve, rather than replace, strong native respiratory evidence. In the MAHNOB-HCI example, many fixed observations are flat, noisy, or phase-inconsistent under the same visual window, and the PARH-OSSM waveform remains limited for the same reason. This is the visual counterpart of Table 2: the model can improve the rate-oriented path, but full waveform recovery still depends on whether the camera-derived observation bank contains recoverable morphology.

Calibration diagnostics

The calibration diagnostics summarise a limited but important boundary. PARH-OSSM stores NIS and Student- t adaptation diagnostics, while OSSM-KF provides comparable stability duration but not the PARH-specific NIS, λ_t , or adaptive reliability

Same-trial waveform topology from full-dataset manifest cases

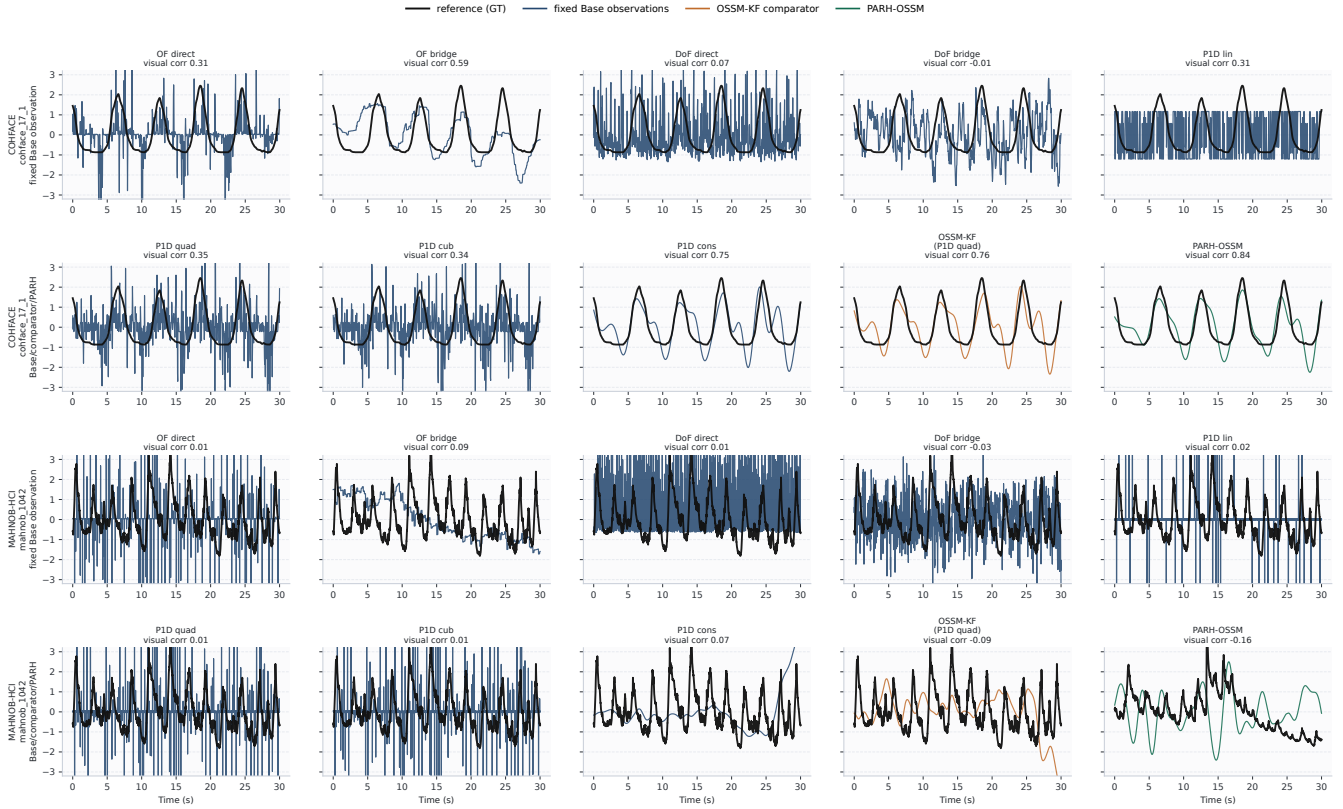


Figure 3. Same-trial all-base waveform overlay atlas for the clean and hard regimes. Each dataset block uses one predefined manifest case from the predefined evaluation cohort and shows all eight fixed Base observation operators, the pre-specified OSSM-KF (PID quad) comparator, and the PARH-OSSM readout against the same reference. This figure is generated from a predefined same-trial manifest. The visual window is chosen by reference visibility only, not by method performance. The atlas should be read together with the strict zero-lag metrics in Table 2; near-flat or noisy fixed-observation traces indicate weak observation topology in that window, not a different reference. Supplementary Figs. S12 and S13 provide observation-median all-class overlay companions.

arrays. MAHNOB-HCI is not explained by gross state-space divergence: NIS remains close to one and Student- t weights do not collapse. The failure is therefore better read as limited respiratory evidence plus reference/observability risk, not as an unstable Kalman-style update. The complete calibration diagnostics are retained in the Supplementary Information.

Ablation evidence and failure decomposition

The rate-source decomposition and Table 3 first ask why the evaluated system does not close MAHNOB-HCI. The rate-source decomposition shows that COHFACE has a strong native timing source and that the reported track is a conservative but still high-correlation readout. In MAHNOB-HCI, however, even the best diagnostic rate source remains in the 3 bpm error range and is only modestly better than the final track. The failure taxonomy shows that this oracle room is itself limited: 400 of 525 MAHNOB trials are bounded by the current observation bank, 19 are likely video/reference limited, 95 have actionable source-selection room, and 9 have oracle room without reliable label-free evidence. Thus the hard regime is not best described as a missing hyperparameter. It is an observability problem: the current fixed OF/DoF/ProfileID sources do not always contain enough respiratory evidence for a deployable adaptive law to recover.

The COHFACE operator-alignment ablation asks a narrow but important question: should respiratory timing and waveform morphology be forced through the same output? The answer is no. A waveform-specialized expert can match aligned morphology, but it loses rate. The decoupled readout preserves the strong oscillatory path while retaining the same aligned waveform CCC and strict CCC. This is the cleanest numeric support for evaluating rate from z_{osc} and waveform morphology from $z_{\text{full}} = z_{\text{osc}} + z_{\text{morph}}$.

The component-level evidence ledger compresses the same story into a component-level diagnostic ledger. It deliberately

Table 3. MAHNOB observability failure taxonomy from the complete-cohort label-free diagnostic analysis. The table should be read as diagnostic evidence, not as a proposed model variant. “Room” is the median difference between the final selected source and the oracle-best source inside the candidate bank.

Category	Trials	Final MAE	Room	Interpretation
Bounded by current observation bank	400/525	2.150	0.290	Current candidate sources have little oracle room; better source selection alone is unlikely to solve these trials.
Source-selection room with posterior evidence	60/525	3.655	1.030	A different source can help and the label-free posterior already contains some usable evidence.
Source-selection room with agreement evidence	35/525	3.960	0.990	A different source can help and cross-source agreement is partly informative.
Likely video/reference limited	19/525	4.510	0.170	Even the best current source remains poor, suggesting weak visual respiratory evidence or reference scale/lag risk.
Oracle room but weak label-free evidence	9/525	5.050	1.940	A better source exists, but current label-free diagnostics do not identify it reliably.
Very low label-free observability	2/525	3.715	0.660	Very low label-free support; retained as explicit hard-observability evidence rather than removed from the evaluation cohort.

separates components that are retained as model structure from components kept only as diagnostics. This prevents the analysis from implying that every explored mechanism is a performance gain while still showing why each retained block exists.

The detailed rate-source decomposition is provided in the Supplementary Information. It shows that the reported MAHNOB-HCI rate track (2.410 bpm) is close to the best diagnostic source available inside the readout bank (2.390 bpm), so the dominant hard-regime failure is not merely the last readout switch.

Table 3 separates several failure mechanisms that would be hidden by a single MAE value. The largest category is bounded by the current observation bank: these trials have little oracle room, so a better last-step selector would not substantially change the result. The two source-selection categories indicate cases where another candidate source could help and where label-free posterior or cross-source agreement evidence already contains partial guidance. The video/reference-limited and very-low observability categories mark trials where the available visual evidence or reference alignment is itself questionable. The “oracle room but weak label-free evidence” category is especially important because it identifies a specific future target: richer label-free observability cues, not merely a more complex state-space smoother.

The operator-alignment and component-ablation ledgers are retained in the Supplementary Information. The most important ablation result is that the decoupled readout preserves the rate-specialized path while matching the waveform-specialized morphology and strict CCC in the COHFACE diagnostic package; a morphology-only expert cannot replace timing-aware estimation. Components without a matched numeric delta remain construction or diagnostic evidence rather than performance-improvement claims.

Mechanism diagnostics and comparison ladder

The mechanism diagnostic table remains COHFACE-focused because the saved per-frame diagnostic arrays are available there. That scope limitation is deliberate rather than hidden, and it separates two questions cleanly: the mechanism table shows how the decomposition routes energy through harmonic, baseline, and residual blocks when the observation is clean and reproducible, while Figure 4 uses the same complete-cohort comparison layer as the main tables to show how fixed operators, the OSSM-KF comparator, and PARH-OSSM move across clean and hard regimes.

The detailed COHFACE mechanism analysis from diagnostic arrays is provided in the Supplementary Information. The main text keeps the graphical comparison ladder because it directly links the fixed observation bank, the comparator, and the PARH-OSSM readout.

Figure 4 is included to keep the comparison visually tied to the observation problem. In COHFACE, several Profile1D-based observations already expose strong respiratory structure, so PARH-OSSM is not presented as a universal replacement for a strong fixed baseline. In MAHNOB-HCI, rate improves while waveform structure remains weak, which is exactly the pattern expected when the limiting factor is the available camera observation rather than a single Kalman-filter parameter. This figure therefore connects the numerical tables to the central interpretation: the framework is useful because it separates timing, form, and failure evidence, not because every metric in every observation regime is forced to improve.

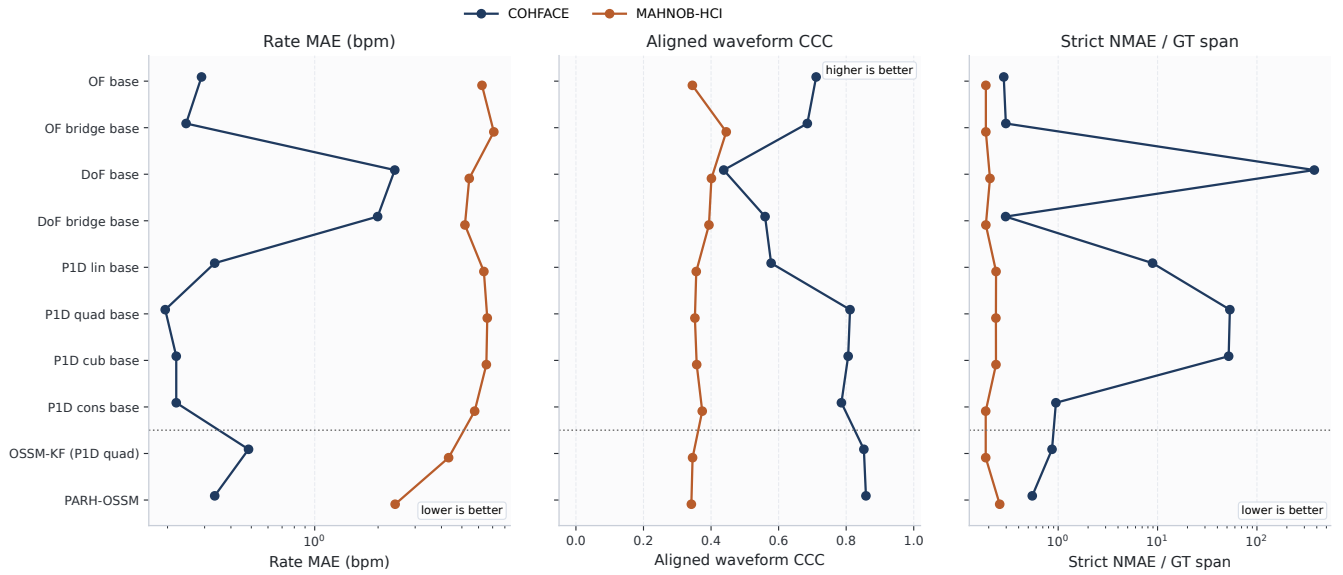


Figure 4. All-base complete-cohort comparison ladder for the eight fixed observation operators, the pre-specified OSSM-KF (P1D quad) comparator, and PARH-OSSM. The left panel shows rate MAE, the middle panel shows aligned waveform CCC, and the right panel shows strict span-normalized error; the dotted divider separates the fixed observation bank from the comparator/proposed readouts. The figure makes explicit that COHFACE is a clean regime where profile evidence is already strong, whereas MAHNOB-HCI remains a hard observability regime even after PARH-OSSM improves the rate track.

Failure-case manifests

Figure 5 makes two of these limitations visible without post hoc cherry-picking because both panels are selected from persistent manifests. The top row shows a residual-heavy PARH case selected without forcing a particular observation class; in the predefined manifest it is an optical-flow-bridge trial. The residual branch is clearly active and the non-oscillatory-need trace rises, but the waveform fit remains poor. This shows the observed limitation directly: the residual path is no longer dormant, yet it is still not a fully identifiable physiological branch that can rescue a strongly mismatched observation. The bottom row shows the worst optical-flow-bridge case from the bridge manifest. This keeps the story honest: the optical-flow bridge is a useful observation route for rate-oriented analysis, but it is not a universal replacement for raw OF. We therefore treat it as an additional observation route rather than collapsing the two.

Figure 5 closes the evidence chain by showing where the current decomposition is informative but incomplete. The residual diagnostic can detect non-oscillatory need, yet a high residual contribution is not automatically a recoverable respiratory waveform when the observation is poor. Likewise, the bridge construction can create a useful rate-oriented cue in some trials, but the worst-case panel shows that it can also fail under the same audited pipeline. These panels therefore prevent the discussion from overstating the method: PARH-OSSM improves interpretability and hard-regime rate tracking, but the current observation bank remains the limiting factor for universal morphology recovery.

Discussion

Why PARH-OSSM gains are still limited

The combined COHFACE and MAHNOB-HCI evidence narrows the problem. The latent decomposition is no longer the dominant failure mode. The present framework has an active second harmonic, non-saturated oscillatory-support scores, usable baseline and residual states, corrected rate routing, and observation-dependent gains that would not appear if the extra states were dormant. The remaining bottleneck is the observation equation and the label-free evidence available to that equation. NIS remains near unity and Student- t weights do not collapse, which reinforces that the hard-regime failures are driven by observation quality rather than gross filter instability.

COHFACE and MAHNOB-HCI expose that bottleneck differently. In clean COHFACE, the integrated system preserves strong rate and aligned morphology. In irregular MAHNOB-HCI, the same framework remains calibrated but the final rate

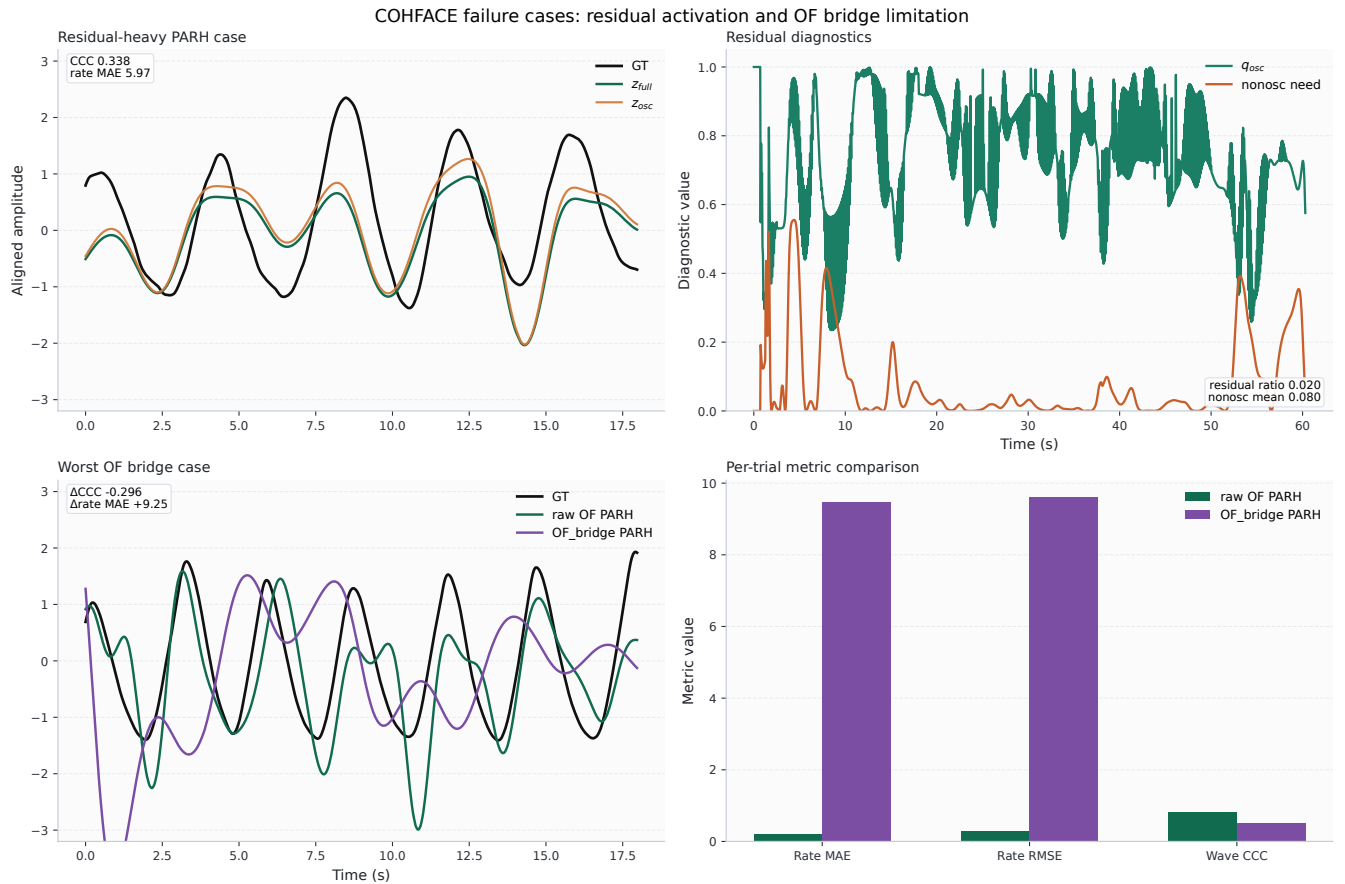


Figure 5. Failure cases selected from predefined manifests. Top row: the residual-heavy PARH case shows elevated residual energy and non-oscillatory need, but still poor waveform fidelity. Bottom row: the worst optical-flow bridge case from the bridge manifest shows that observation construction is not uniformly beneficial. Together these panels bound the claims: the model is more interpretable than a generic robust filter, but the observation and residual branches are still incomplete.

correlation and waveform fidelity are weak. This means the proposed state-space organization is not enough by itself: a hard target regime also needs richer or more reliable respiratory observations.

What the evidence supports

Three claims are now defensible.

- Observation construction matters. Raw optical flow and the optical-flow bridge come from the same source but lead to materially different rate-versus-waveform trade-offs.
- Observation roles are now identifiable. Quadratic and cubic Profile1D routes are waveform-primary harmonic cues, the optical-flow bridge is a rate-oriented constructed cue, Profile1D consensus adds rate-supporting agreement evidence, and raw/bridged depth-of-flow routes remain auxiliary cues.
- The hard-regime failure is measurable rather than opaque. MAHNOB-HCI does not support strong waveform claims under the current observation bank, but the observability taxonomy identifies when the bank has little oracle room and when source-selection evidence exists but remains insufficient for deployable source routing.

Current boundaries and next observation requirements

Several boundaries still separate the present system from a fully closed adaptive multi-observation state-space model.

- PARH-OSSM is an integrated but bounded observation-law system. It is not a fully general time-local multi-observation matrix $H_{m,t}$ with explicit source covariance for every observation.
- The strict reconstruction tables show that aligned morphology is not yet the same as true waveform recovery. The present framework can line up breath shape after normalisation and lag correction, but it still fails badly when gain, offset, and zero-lag timing must all be correct at once.

- The present label-free reliability features are not sufficient for deployment-style routing. They can reject some within-source consistency patterns, but they still confuse repeated bridge-like waveforms with genuinely recoverable respiratory morphology on difficult MAHNOB-HCI trials.
- Comparator symmetry is incomplete because OSSM-KF is intentionally a stripped-down resonator-plus-Kalman comparator. It shares the broad state-space inference machinery, but it does not save the same per-frame calibration arrays, posterior source roles, or posterior frequency uncertainty fields as PARH-OSSM.
- The residual branch is more interpretable than before, but it is still a diagnosed leftover pathway rather than a separately validated physiological event branch.
- Multi-observation fusion is only partially closed. The integrated readout uses bounded integration and explicit source analyses, but the next-generation model should make every source's role, covariance, and low-observability decision explicit inside one observation equation.

This boundary also clarifies how the framework should interact with rPPG and deep learning. rPPG-derived respiratory modulation, learned motion fields, or learned ROI-quality maps are natural additional observation operators, not conceptual competitors to the state/readout decomposition^{30,31}. A future learned front end should be judged by whether it produces label-free respiratory evidence and uncertainty that can enter the observation law without using target labels at test time. Conversely, the state-space layer should keep the timing, morphology, baseline, residual, calibration, and low-observability diagnostics explicit, so that a deep model does not simply hide the same failure modes inside an uninterpretable waveform output.

Limitations and next steps

The current analysis no longer has an incomplete MAHNOB-HCI cohort limitation: both major datasets are evaluated over their predefined cohorts. The remaining limitations are scientific.

First, the integrated system is strong on COHFACE but still weak on MAHNOB-HCI. That gap should not be explained away as a formatting artifact: MAHNOB rate correlation is low, aligned waveform CCC is modest, and strict waveform CCC is 0.000 in the complete-cohort table. Second, the calibration diagnostics do not show gross state divergence, so simply retuning a Kalman parameter is unlikely to be the right scientific fix. Third, the label-free observability analysis shows that many MAHNOB trials have little recoverable room inside the present OF/DoF/Profile1D bank. The conclusion is that the framework is useful and auditable, but the present observation bank is not sufficient for universal camera-based respiratory reconstruction.

The next development stage should therefore prioritise four tasks. (1) Build richer respiratory observations before adding more output-side gates, especially phase-anchored morphology cues that survive person and environment shift. (2) Make label-free reliability time-local, so trust can change within a recording instead of being inferred mostly at the clip level. (3) Make the residual branch identifiable as respiratory residual structure rather than a catch-all observation-error absorber. (4) Promote multi-observation fusion only when each source carries an explicit role, covariance, and low-observability flag; naive source stacking would obscure the main lesson of the present analysis.

Data availability

COHFACE and MAHNOB-HCI are public research datasets available from their respective providers under their original access terms. V4V and SCAMPS are used only as external rate-only and synthetic-control scope checks, respectively, and are not mixed into the real-data waveform main metrics. The metric tables, figure manifests, and diagnostic summaries used in this study are included with the release artifacts and can be regenerated from the command ledger once the raw datasets have been obtained from their providers. Raw third-party video and physiology files are not redistributed.

Code availability

The code base contains the PARH-OSSM implementation, fixed observation operators, figure/table generation scripts, and verification scripts used to produce the reported analyses. The public release includes the command ledger `execute.md`, configuration files, table-ready CSVs, figure manifests, environment/setup notes, and audit scripts used to reproduce the reported values. The code and derived artifacts are available at <https://github.com/GongJae00/resPyre>. The paper-submission snapshot is tagged as `v1.0.0-paper-submission`; a DOI-minting archival release will be created before publication.

Ethics declarations

This work uses previously released public or research datasets and does not introduce new human-subject data collection. Dataset-specific consent and ethics conditions remain governed by the original dataset releases. The present analysis reports aggregate algorithmic performance and diagnostic summaries.

References

1. Bartula, M., Tigges, T. & Muehlsteff, J. Camera-based system for contactless monitoring of respiration. *Proc. IEEE Eng. Medicine Biol. Soc. (EMBC)* 2672–2675 (2013).
2. Tarassenko, L. *et al.* Non-contact video-based vital sign monitoring using ambient light and auto-regressive models. *Physiol. Meas.* **35**, 807–831 (2014).
3. Janssen, R., Wang, W., Moço, A. & de Haan, G. Video-based respiration monitoring with automatic region of interest detection. *Physiol. Meas.* **37**, 100–114 (2016).
4. Li, M. H., Yadollahi, A. & Taati, B. A non-contact vision-based system for respiratory rate estimation. *Proc. IEEE Eng. Medicine Biol. Soc. (EMBC)* 2119–2122 (2014).
5. Bai, H., Wang, Y. & Song, S. Camera-based respiratory rate monitoring from thoracic movement. *Comput. Biol. Medicine* **130**, 104186 (2021).
6. Welch, P. D. The use of fast Fourier transform for the estimation of power spectra. *IEEE Transactions on Audio Electroacoustics* **15**, 70–73 (1967).
7. Boccignone, G. *et al.* Remote respiration measurement with rgb cameras: A review and benchmark. *ACM Comput. Surv.* **58**, 114:1–114:36, DOI: [10.1145/3771763](https://doi.org/10.1145/3771763) (2025).
8. Wang, W. & den Brinker, A. C. Algorithmic principles of camera-based respiratory motion extraction. *arXiv preprint arXiv:2105.07537* (2021).
9. Harvey, A. C. *Forecasting, Structural Time Series Models and the Kalman Filter* (Cambridge University Press, 1989).
10. Durbin, J. & Koopman, S. J. *Time Series Analysis by State Space Methods* (Oxford University Press, 2012), 2 edn.
11. Särkkä, S. *Bayesian Filtering and Smoothing* (Cambridge University Press, 2013).
12. Mehra, R. K. On the identification of variances and adaptive Kalman filtering. *IEEE Transactions on Autom. Control.* **15**, 175–184 (1970).
13. Mohamed, A. H. & Schwarz, K. P. Adaptive Kalman filtering for INS/GPS. *J. Geod.* **73**, 193–203 (1999).
14. Piché, R., Särkkä, S. & Kannala, J. Recursive outlier-robust filtering and smoothing for nonlinear systems using the multivariate student's *t* distribution. In *IEEE International Workshop on Machine Learning for Signal Processing*, 1–6 (2012).
15. Roth, M., Özkan, E. & Särkkä, S. A student's *t* filter for heavy tailed process and measurement noise. *Proc. IEEE Int. Conf. on Acoust. Speech Signal Process. (ICASSP)* 5770–5774 (2013).
16. Huang, Y., Zhang, Y., Zhao, Y. & Chambers, J. A. A novel robust student's *t*-based Kalman filter. *IEEE Transactions on Aerosp. Electron. Syst.* **53**, 1545–1554 (2017).
17. Särkkä, S. *et al.* Dynamic retrospective filtering of physiological noise in BOLD fMRI: DRIFTER. *NeuroImage* **60**, 1517–1527 (2012).
18. Beck, A. M., He, M. & Purdon, P. L. State space oscillator models for neural/electrophysiological data. *Proc. IEEE Eng. Medicine Biol. Soc. (EMBC)* (2018).
19. Solin, A. & Särkkä, S. Explicit link between periodic covariance functions and state space models. In *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Statistics*, vol. 33 of *Proceedings of Machine Learning Research*, 904–912 (2014).
20. Chaichulee, S., Villarroel, M., Jorge, J., Rosenberger, C. & Tarassenko, L. Multi-task convolutional neural network for patient monitoring under camera. *Proc. IEEE Int. Jt. Conf. on Neural Networks (IJCNN)* 2094–2101 (2017).
21. Perepelkina, O., Artemyev, M., Churikova, M. & Grinenko, M. Hearttrack: Convolutional neural network for remote video-based heart rate monitoring. *Proc. IEEE/CVF Conf. on Comput. Vis. Pattern Recognit. Work.* 288–289 (2020).
22. Oku, Y. Temporal variations in the pattern of breathing: Techniques, sources, and applications to translational sciences. *The J. Physiol. Sci.* **72**, 22 (2022).
23. van den Bosch, O. F. C., Alvarez-Jimenez, R., de Grooth, H.-J., Girbes, A. R. J. & Loer, S. A. Breathing variability—implications for anaesthesiology and intensive care. *Critical Care* **25**, 280 (2021).
24. Kalman, R. E. A new approach to linear filtering and prediction problems. *Transactions ASME—Journal Basic Eng.* **82**, 35–45 (1960).
25. Rauch, H. E., Tung, F. & Striebel, C. T. Maximum likelihood estimates of linear dynamic systems. *AIAA J.* **3**, 1445–1450 (1965).
26. Farnebäck, G. Two-frame motion estimation based on polynomial expansion. *Proc. Scand. Conf. on Image Analysis* 363–370 (2003).
27. Heusch, G., Anjos, A. & Marcel, S. A reproducible study on remote heart rate measurement. In *arXiv:1709.00962* (2017).
28. Soleymani, M., Lichtenauer, J., Pun, T. & Pantic, M. A multimodal database for affect recognition and implicit tagging. *IEEE Transactions on Affect. Comput.* **3**, 42–55 (2012).
29. Lin, L. I.-K. A concordance correlation coefficient to evaluate reproducibility. *Biometrics* **45**, 255–268 (1989).

30. Gwak, M. *et al.* Multimodal breathing rate estimation using facial motion and RPPG from RGB camera. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, DOI: [10.1109/ICASSP48485.2024.10446086](https://doi.org/10.1109/ICASSP48485.2024.10446086) (2024).
31. Du, J., Liu, S.-Q., Zhang, B. & Yuen, P. C. Weakly supervised rPPG estimation for respiratory rate estimation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2391–2397, DOI: [10.1109/ICCVW54120.2021.00271](https://doi.org/10.1109/ICCVW54120.2021.00271) (2021).

Acknowledgements

This work was supported by the authors' institutions.

Author contributions statement

Y.-J.C. implemented PARH-OSSM, performed the reproducibility checks, analysed the trial-level diagnostic results, and prepared the manuscript. D.-Y.K. supervised the project and reviewed the manuscript direction and interpretation.

Additional information The authors declare no competing interests.